

TrustLedger SRM Intelligence Architecture Review — Part 3

Intelligence Architecture & Traceability Model

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Review status

Part 3 translates the Core Intelligence Model and Capability Map into a conceptual architecture for how TrustLedger should create, preserve, explain and evolve intelligence. It is a design reference, not an implementation instruction.

1. Purpose of Part 3

Part 1 established the intelligence cycle. Part 2 established the boundary of what TrustLedger should own. Part 3 addresses the next question:

How should intelligence be structured so that different SRM methodologies, signals, models and future capabilities can operate against the same trusted foundation?

The answer proposed here is a layered, evidence-linked intelligence architecture with explicit provenance, context, interpretation, uncertainty, human governance and learning.

2. Architectural Proposition

TrustLedger should separate operational facts from contextual information, analytical signals, interpretations, intelligence, recommendations and human decisions.

This separation is essential because these things have different levels of certainty, different sources, different governance requirements and different lifecycles.

3. The Intelligence Stack

Operational Spine

Organisation → Project / Programme → Place / Context → Stakeholder → Relationship → Engagement → Commitment → Issue / Grievance → Evidence → Decision → Action → Outcome → Learning

Intelligence Stack

Evidence → Context → Signal → Interpretation → Intelligence → Recommendation → Human Decision → Action → Outcome → Learning

Extensible Services

Methodologies • Rules • Models • Indicators • AI Assistance • External Data • Integrations

4. The Fundamental Separation

TrustLedger should maintain a clear distinction between:

Fact: something recorded or verified.

Observation: something reported, witnessed or captured from a source.

Signal: a meaningful pattern or change detected from available information.

Interpretation: a plausible explanation of the signal.

Intelligence: a structured, decision-useful synthesis of evidence and context.

Recommendation: a proposed response.

Decision: the accountable human choice.

Action: what is actually done.

Outcome: what changed as a result.

Learning: what should influence future practice.

5. Why This Separation Matters

Without these distinctions, TrustLedger risks converting assumptions into facts, scores into truth, recommendations into decisions and activity into impact.

The architecture should make it possible to answer: **What do we know? How do we know it? What does it suggest? What else could explain it? Who interpreted it? What was decided? What happened afterwards?**

6. Intelligence Object — Conceptual Definition

A future TrustLedger intelligence object should conceptually contain:

Subject: stakeholder, relationship, engagement, project, place, issue, commitment or community context.

Signal: the condition or pattern detected.

Evidence: supporting observations and records.

Context: relevant circumstances.

Interpretation: explanation or hypothesis.

Method: rule, framework, model or professional methodology used.

Confidence: indication of evidential strength or uncertainty.

Time: when the observation, signal and interpretation apply.

Recommendation: proposed response where appropriate.

Governance: who reviewed, accepted, rejected or modified it.

Outcome: subsequent result when available.

7. Evidence Provenance

Every material intelligence statement should be traceable back to its supporting evidence wherever technically and practically possible.

Provenance should identify the source, capture date, relevant record, methodology or rule, transformations applied and human review where applicable.

No evidence trail → lower trust in the intelligence.

8. Context Is a First-Class Architectural Element

Context should not be buried inside notes or narrative fields. It should be capable of being associated with the intelligence being produced.

Context may include project stage, place, community conditions, stakeholder operating context, BAU pressure, capacity, competing priorities, political/institutional circumstances, previous engagements, unresolved commitments and related issues.

9. Temporal Intelligence

Stakeholder relationships change. Therefore, intelligence should be understood as time-dependent rather than permanently true.

The architecture should distinguish the time of an observation from the time at which an interpretation or assessment was made. Historical states should remain recoverable where appropriate.

This prevents a stakeholder's historical condition from being treated as their permanent identity.

10. Relationship-Centred Intelligence

Stakeholder intelligence should not be isolated from relationships. A stakeholder's position may depend on relationships with other stakeholders, institutions, communities, project teams and decision-makers.

The architecture should therefore support relationship references without forcing an immediate graph database or visual network product.

11. Influence, Impact, Capacity & Commitment

The four dimensions identified during the architecture review — **Influence × Impact × Capacity × Commitment** — should be treated as contextual intelligence dimensions rather than permanent labels.

Additional dimensions such as BAU pressure, availability, competing priorities, accountability clarity and engagement burden can extend the operating-context model without changing the stakeholder's core identity.

12. Signal Architecture

Signals should be generated through multiple possible mechanisms:

- rule-based conditions
- threshold checks
- trend detection
- pattern recognition
- professional methodologies
- comparative analysis
- human observations
- external data
- AI-assisted analysis

The source mechanism should remain identifiable. TrustLedger should not present every signal as if it came from the same analytical method.

13. Interpretation Architecture

Interpretation should support more than one plausible explanation when evidence is ambiguous.

Example: declining participation may indicate resistance, consultation fatigue, capacity constraints, poor timing, inadequate accessibility, competing priorities, mistrust or perceived irrelevance.

The architecture should preserve hypotheses rather than prematurely collapse uncertainty.

14. Confidence and Uncertainty

Confidence should not be confused with probability or certainty unless a validated methodology explicitly defines such a measure.

A useful conceptual vocabulary is: **verified, strongly supported, indicative, emerging, uncertain, disputed, insufficient evidence.**

The precise taxonomy should be determined later through methodology and product review.

15. Methodology Registry Concept

Future TrustLedger architecture should be able to identify which professional methodology, framework, rule set or analytical model produced an intelligence result.

A methodology definition should conceptually include: name, purpose, version, author/owner, required inputs, analytical logic, outputs, evidence requirements, limitations and governance status.

This creates methodological transparency and allows multiple approaches to coexist.

16. Rule and Model Separation

Rules and models should not be confused with the underlying intelligence records.

Rule: explicit condition or logic.

Model: analytical mechanism that transforms inputs into an output.

Methodology: professional framework governing how an assessment is conducted and interpreted.

Intelligence: the resulting decision-useful interpretation.

This separation makes future evolution safer.

17. AI Assistance Layer

AI should operate as an optional assistance layer around the intelligence architecture. It may assist with extraction, summarisation, classification, drafting, pattern exploration or alternative interpretations.

AI-generated outputs should retain source references and should be distinguishable from verified records and human decisions.

AI must not silently write authoritative stakeholder truth.

18. Human Governance Layer

Human governance should be explicit wherever intelligence can materially influence stakeholder treatment, escalation, engagement strategy, risk response or reporting.

The architecture should support states such as **suggested** → **reviewed** → **accepted / modified / rejected** → **acted upon** → **outcome recorded**.

19. Recommendation Architecture

Recommendations should be separate from decisions. A recommendation may include: issue; evidence; context; rationale; alternative explanations; proposed response; urgency; responsible role; and supporting methodology.

This allows different recommendations to be compared and human judgement to remain visible.

20. Outcome and Learning Architecture

The architecture should eventually connect decisions and actions to outcomes. This is essential if TrustLedger is to become a learning system rather than merely a monitoring system.

Learning may identify which engagement approaches worked, which assumptions proved wrong, which risks were correctly or incorrectly interpreted, and which stakeholder conditions changed over time.

21. Cross-Domain Intelligence

Intelligence should be able to draw together records from multiple domains without duplicating them.

For example, a stakeholder-capacity warning might combine stakeholder operating context, engagement attendance, commitment delays, project timing and relationship signals. The resulting intelligence should reference those underlying records rather than copy them into a new isolated data store.

22. Intelligence Graph — Conceptual, Not Prescriptive

Conceptually, TrustLedger intelligence forms a graph:

Stakeholder ↔ Relationship ↔ Engagement ↔ Commitment ↔ Issue ↔ Evidence ↔ Place ↔ Project
with **Signals → Interpretations → Intelligence → Recommendations → Decisions → Actions → Outcomes**
connected to the relevant records.

This does **not** mean TrustLedger should immediately adopt graph-database technology. The graph is a conceptual relationship model, not an implementation mandate.

23. Architecture for Future Capability

A new intelligence capability should be able to reuse existing evidence, context, stakeholder, relationship, engagement, project and outcome structures.

If a future methodology requires a new assessment dimension, it should preferably add an extensible analytical definition rather than force redesign of the operational spine.

24. Intelligence Lifecycle

Capture → Validate → Contextualise → Detect Signal → Interpret → Produce Intelligence → Review → Decide → Act → Observe Outcome → Learn → Reassess

The lifecycle should permit intelligence to be revised as new evidence appears. Historical intelligence should not simply be overwritten.

25. Architectural Guardrails

Guardrail 1: Never make an interpretation indistinguishable from a fact.

Guardrail 2: Never lose the evidence trail for material intelligence.

Guardrail 3: Never make a methodology inseparable from the core data model unless formally justified.

Guardrail 4: Never treat a score as an objective truth without its methodology and context.

Guardrail 5: Never allow AI output to become authoritative without human governance.

Guardrail 6: Never design intelligence so that a future methodology requires core redesign by default.

Guardrail 7: Never confuse activity completion with outcome or impact.

26. Preliminary Architectural Pattern

CORE DATA



EVIDENCE + CONTEXT



SIGNAL SERVICES



METHODS / RULES / MODELS



INTELLIGENCE SERVICES



RECOMMENDATION



HUMAN GOVERNANCE



DECISION + ACTION



OUTCOME + LEARNING

AI and external data operate as controlled supporting services across the appropriate layers.

27. What Part 3 Does Not Authorise

This architecture does not authorise a new intelligence database, graph database, intelligence DocTypes, methodology registry implementation, predictive engine, AI architecture, new dashboards, new scoring system, automatic decision-making, or migration of existing TrustLedger structures.

Those are implementation questions for later, after the architecture is formally approved.

28. Part 3 Conclusion

The intelligence architecture should be more stable than any individual intelligence capability.

TrustLedger should preserve a durable separation between evidence, context, signals, interpretations, intelligence, recommendations, decisions, actions, outcomes and learning. Professional methodologies, rules, models, AI services and external data should operate around this stable structure.

The architecture should make new intelligence possible without making old intelligence fragile.

This establishes the foundation for **Part 4: TrustLedger Methodology & Framework Architecture** — determining how professional SRM methodologies can be introduced, versioned, compared, configured and retired without locking TrustLedger into a single professional doctrine.

Appendix A — Conceptual Intelligence Architecture

CORE OPERATIONAL SPINE

Organisation → Project → Place/Context → Stakeholder → Relationship → Engagement → Commitment
→ Issue → Evidence → Decision → Action → Outcome → Learning

INTELLIGENCE LAYERS

Evidence + Context → Signal → Interpretation → Intelligence → Recommendation → Human Decision →
Action → Outcome → Learning

EXTENSIBLE SERVICES

Methodologies • Rules • Models • Indicators • AI Assistance • External Data • Integrations

Appendix B — Minimum Traceability Questions

Question	Architectural requirement
What is being assessed?	Explicit subject and scope.
What evidence supports it?	Traceable source records.
What context applies?	Relevant temporal, project, place and stakeholder context.
What signal was detected?	Explicit distinction from interpretation.
How was it interpreted?	Method, rule, model or human reasoning identified.
How certain is it?	Confidence/uncertainty represented without false precision.
What was recommended?	Recommendation separate from decision.
Who decided?	Human accountability recorded where material.
What happened?	Action and outcome traceability.
What was learned?	Feedback into future intelligence.

Architectural boundary: This document is strategic reference guidance. It does not authorise implementation.